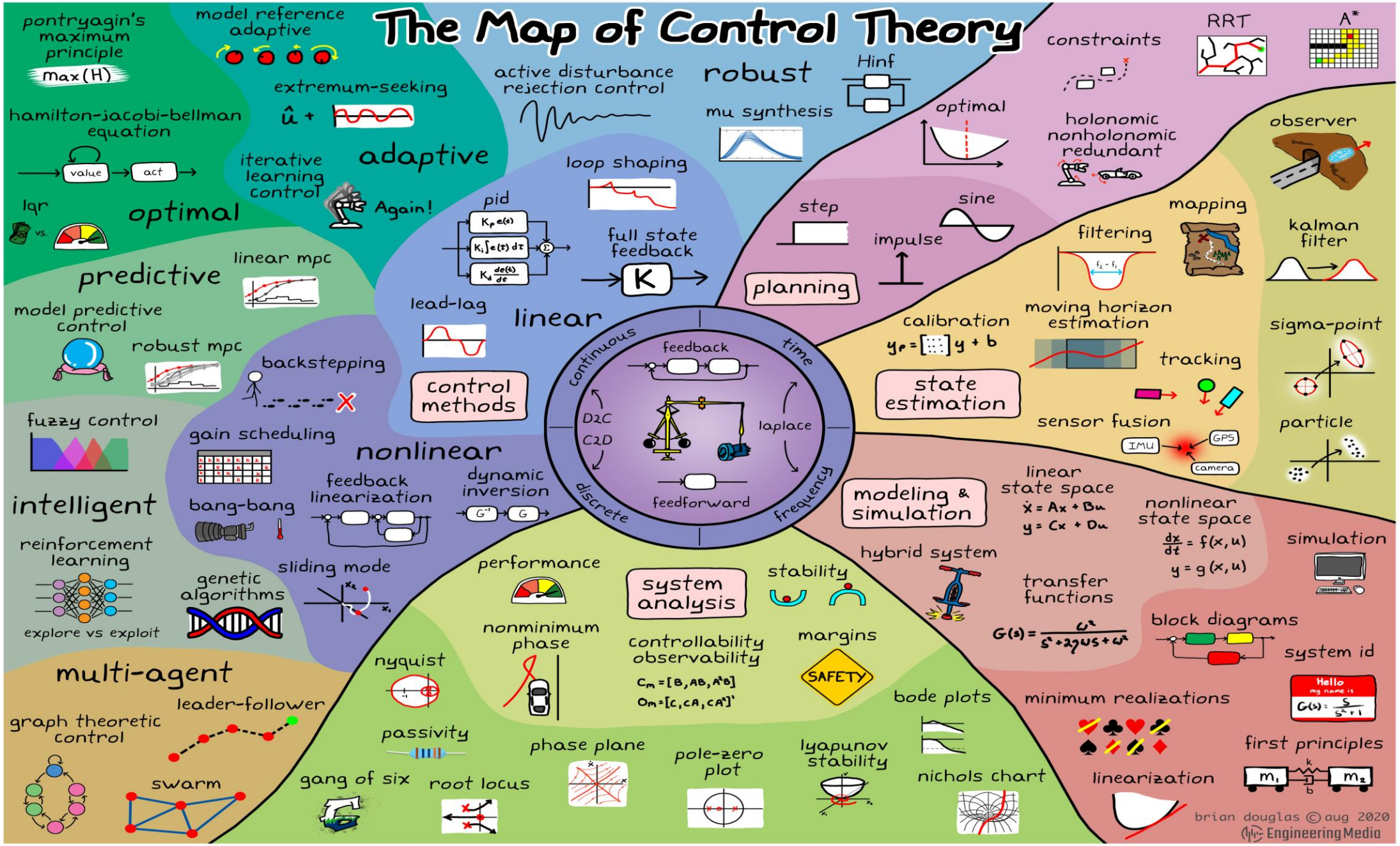
MAS508-G Control Theory
Lecture 2: System Modelling

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Lecture 2b: State-Space Representation

Book “Control Systems Engineering”, Chapter 3


The Map of Control Theory
pontryagin's maximum principle
Max(H)
model reference adaptive
extremum-seeking
active disturbance rejection control
robust
Hinf
mu synthesis
constraints
RRT
A*
hamilton-jacobi-bellman equation
value
act
iterative learning control
adaptive
Again!
loop shaping
optimal
holonomic nonholonomic redundant
observer
lqr
vs.
optimal
pid
step
sine
impulse
full state feedback
K
planning
mapping
filtering
kalman filter
predictive
linear mpc
model predictive control
robust mpc
lead-lag
linear
control methods
calibration
moving horizon estimation
sigma-point
backstepping
state estimation
tracking
fuzzy control
gain scheduling
nonlinear
sensor fusion
IMU
GPS
camera
particle
continuous
discrete
time
frequency
feedback
feedforward
laplace
D2C
C2D
intelligent
feedback linearization
dynamic inversion
bang-bang
modeling & simulation
linear state space
nonlinear state space
simulation
reinforcement learning
explore vs exploit
genetic algorithms
sliding mode
hybrid system
transfer functions
block diagrams
system id
performance
system analysis
stability
nonminimum phase
controllability observability
margins
SAFETY
nyquist
bode plots
minimum realizations
Hello
multi-agent
leader-follower
graph theoretic control
swarm
passivity
phase plane
pole-zero plot
lyapunov stability
nichols chart
first principles
gang of six
root locus
linearization
brian douglas © aug 2020
Engineering Media

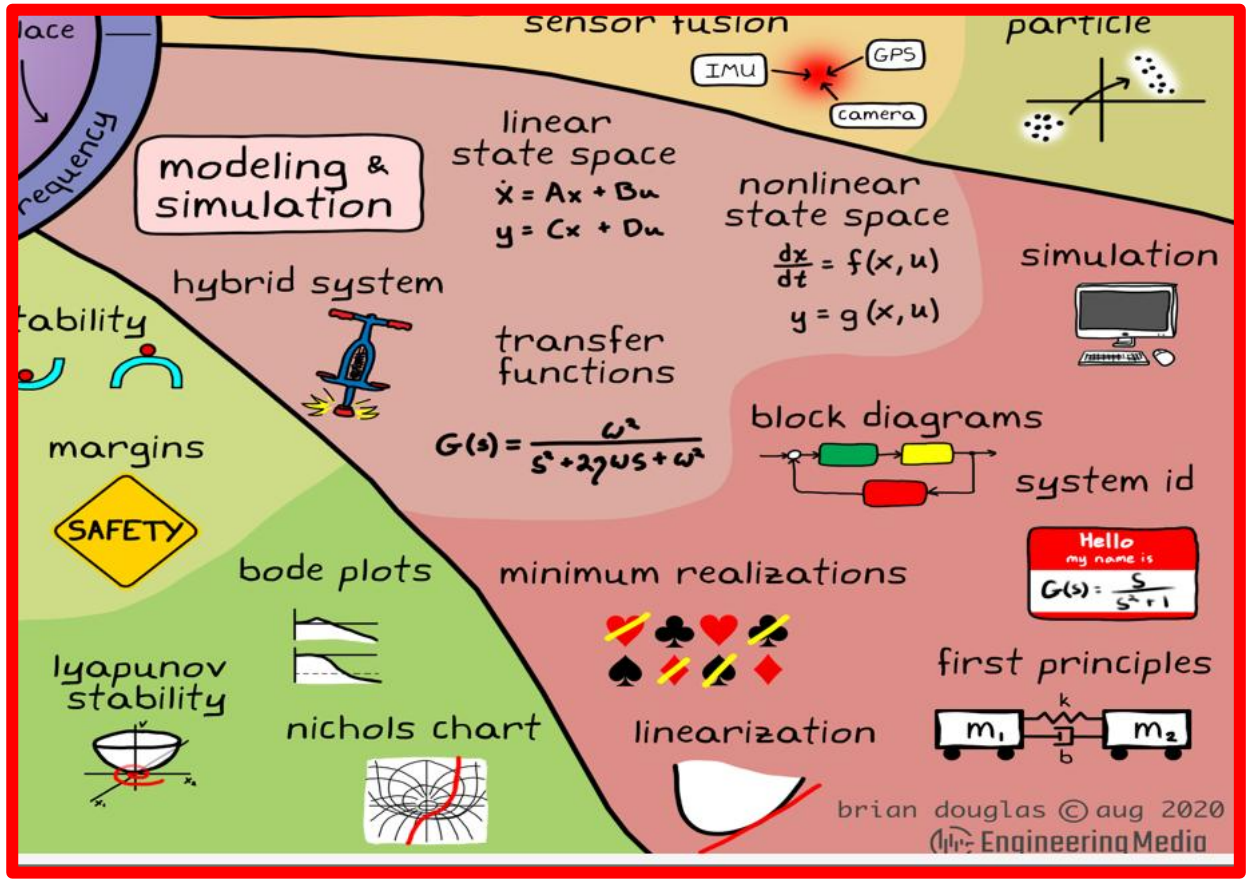
Lecture 2b State-Space Representation

a) Review of Theoretical Modeling
   - Differential equations
   - Transfer functions
   - Block diagrams
   - Frequency response

b) State-Space Representation

c) Experimental Modeling

d) Linearization



What is state-space?

Uses state variables to describe a set of differential equations

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Why model in state-space?

- Easier for more complex systems with multiple degrees of freedom (DOFs), i.e. MIMO systems
- Many control design techniques are available for systems in the state-space form



Classic Control vs. Modern Control

"Classic" Control

- Uses transfer functions to model systems
- Typically for Single-Input Single-Output (SISO) systems
- Uses techniques like frequency-based design and root locus

"Modern" Control

- Uses state-space variable to model system
- Good for Multiple-Input Multiple Output (MIMO) systems
- Represents multiple elements of systems

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

State-Space Representation

General form:

$$\dot{x} = f(x, u)$$
$$y = h(x, u)$$

Where:

x = vector of state variables $(n \times 1)$

u = the control input vector $(r \times 1)$

y = is the output vector $(m \times 1)$

This is for a **n^{th} order system with r inputs and m outputs.**

This can describe both *linear* and *nonlinear* systems.

State-Space: Linear system

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Where:

A = system matrix $(n \times n)$

B = input matrix $(n \times r)$

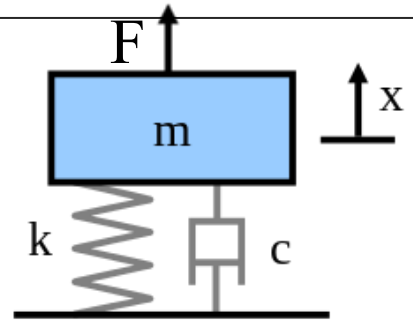
C = output matrix $(m \times n)$

D = feedforward matrix $(m \times r)$

The system state

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

Modelling Terminology

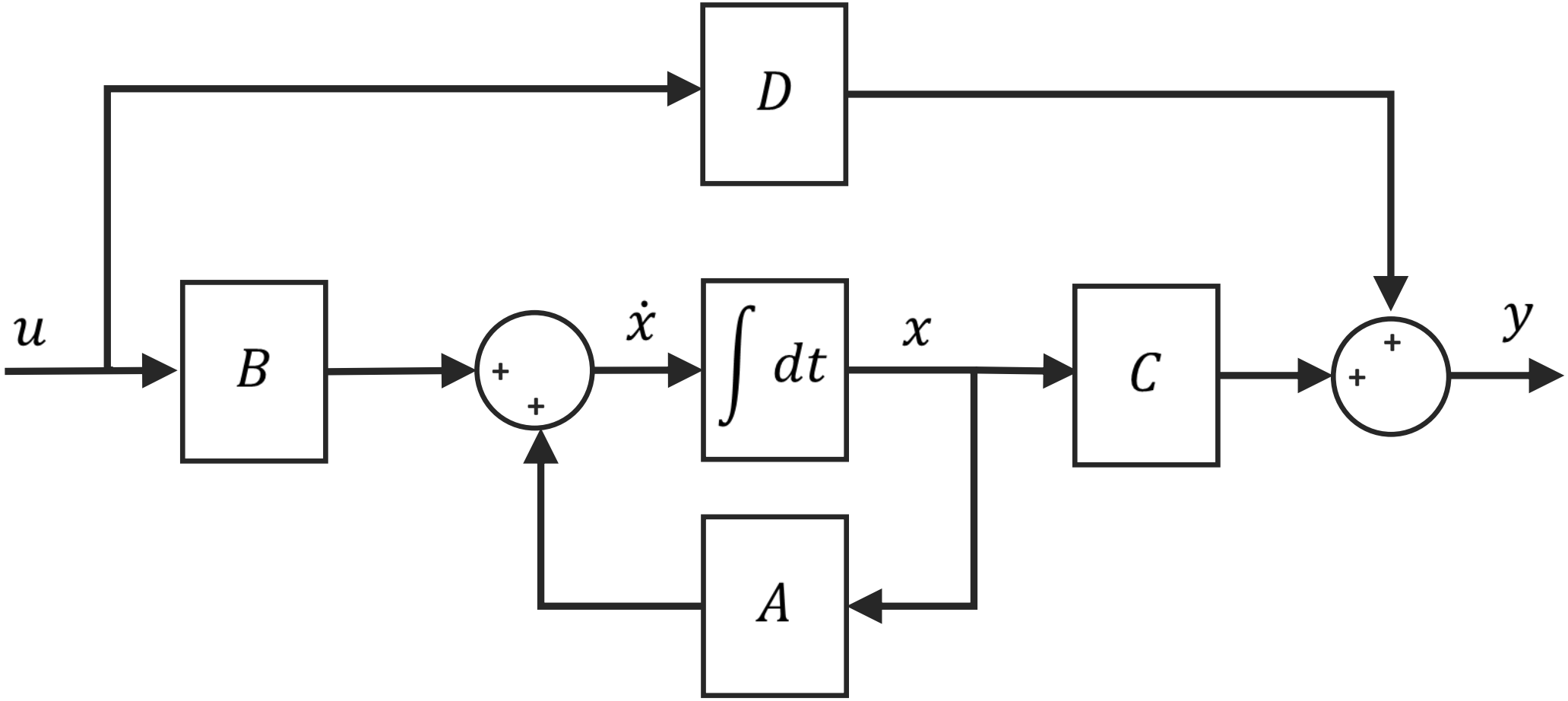


- State captures effects of the past
- Inputs describe external excitation
- Dynamics describes how the state changes
  - Based on current state and inputs
- Outputs describe measured quantities
  - Function of state and inputs
  - Often subset of state

- Example: mass-spring-damper system
  - State: position and velocity of mass $(x,\dot{x})$
  - Input: force input $(u=F)$
  - Dynamics: based on mass (m), stiffness (k) and damping (c) of system
  - Output: measured position of mass $(y=x)$

State-Space Block Diagram

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$



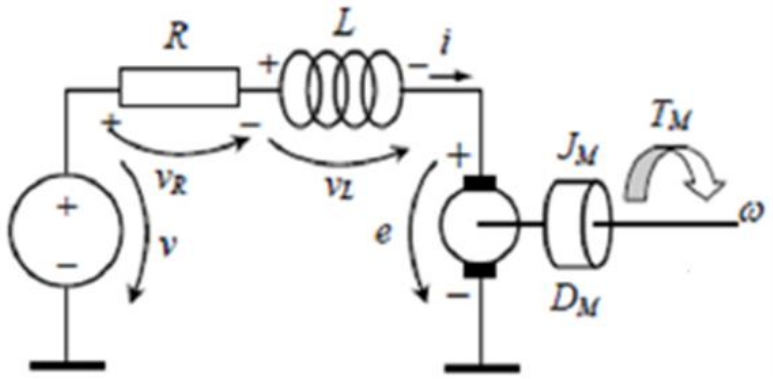
Modelling Procedure

1. Find differential equations of a system
2. Define system state x, input u, and output y
3. Write dynamics of system as a system of a set of first-order equations

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

- State space representation is not unique.

Example: State-Space of DC Motor





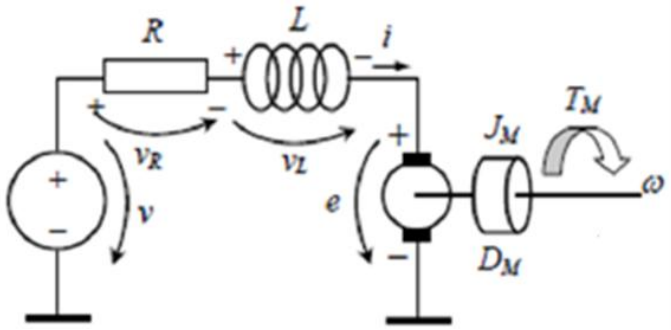
- **Equations of motions:**

$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - K_e\omega(t) = 0$$

$$\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = K_T i(t)$$

- Look at examples with different selection of states, inputs and outputs.
- State space representation is not unique.

Example 1: State-Space of DC Motor





- **Equations of motions:**

$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - K_e\omega(t) = 0$$

$$\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = K_T i(t)$$

- Defining the state variable as:

$$x(t) = [\omega(t) \quad \dot{\omega}(t)\ ]^T$$

- Input: Applied voltage

$$u(t) = v(t)$$

- Output: Motor speed

$$y(t) = \omega(t)$$

Example 1: DC Motor

Equations

$$v(t) - R\, i(t) - L\frac{di(t)}{dt} - K_e\omega(t) = 0$$

$$J_M\frac{d\omega(t)}{dt} = K_T\, i(t) + D_M\omega(t) \longrightarrow i(t) = \frac{1}{K_T}\left(J_M\,\dot{\omega}(t) - D_M\omega(t)\right)$$

$$v(t) - R\frac{1}{K_T}\left(J_M\,\dot{\omega}(t) - D_M\omega(t)\right) - L\frac{1}{K_T}\left(J_M\,\ddot{\omega}(t) - D_M\,\dot{\omega}(t)\right) - k_e\omega(t) = 0$$

$$\frac{L\,J_M}{K_T}\ddot{\omega}(t) = v(t) - \frac{1}{K_T}(RJ_M - LD_M)\dot{\omega}(t) - \left(K_e - \frac{1}{K_T}RD_M\right)\omega(t)$$

$$\ddot{\omega}(t) = -\frac{1}{L\,J_M}(K_TK_e - RD_M)\omega(t) - \frac{1}{L\,J_M}(RJ_M - LD_M)\dot{\omega}(t) + \frac{K_T}{L\,J_M}v(t)$$

Example 1: DC Motor State-Space Model

State vector: $x = [x_1 \;\; x_2]^T = [\omega(t) \;\; \dot{\omega}(t)]^T$
Input: $u = v(t)$
Output: $y = \omega(t) = x_1$

Equations:

$$\dot{x}_1(t) = \dot{\omega}(t) = x_2(t)$$

$$\dot{x}_2(t) = \ddot{\omega}(t) = -\frac{1}{L\,J_M}(K_T K_e - RD_M)x_1(t) - \frac{1}{L\,J_M}(RJ_M - LD_M)x_2(t) + \frac{K_T}{L\,J_M}u(t)$$

State-Space: $\dot{x} = Ax + Bu$

$$y = Cx + Du$$

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{(K_T K_e - RD_M)}{L\,J_M} & -\frac{(RJ_M - LD_M)}{L\,J_M} \end{bmatrix},\; B = \begin{bmatrix} 0 \\ \frac{K_T}{LJ_M} \end{bmatrix},\quad C = [1 \quad 0],\quad D = 0$$

Example 2: State-Space of DC Motor

- States: the current $i(t)$ and the rotational speed $\omega(t)$
- Input: the voltage source $v(t)$
- The output is the rotational speed $\omega(t)$

$$\frac{di(t)}{dt} = \frac{1}{L}\left(-R\, i(t) - K_e\omega(t) + v(t)\right)$$

$$\frac{d\omega(t)}{dt} = \frac{1}{J_M}\left(K_T\, i(t) + D_M\omega(t)\right)$$

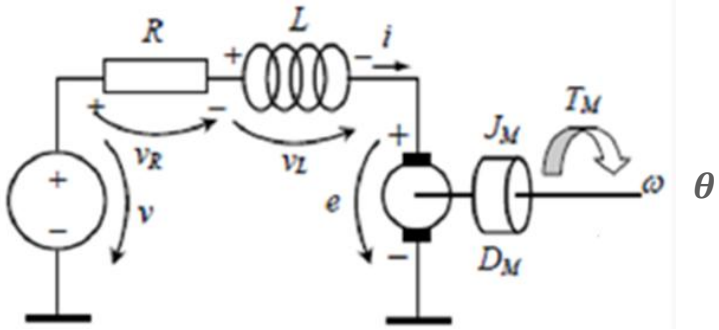
State: $x = [x_1 \;\; x_2]^T = [i(t) \;\; \omega(t)]^T$
Input: $u = v(t)$
Output: $y = \omega(t) = x_2$

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$$A = \begin{bmatrix} -\frac{R}{L} & -\frac{K_e}{L} \\ \frac{K_T}{J_M} & \frac{D_M}{J_M} \end{bmatrix}, \qquad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$$

$$C = [0 \;\; 1], \;\; D = 0$$

Example 3: State-Space of DC Motor





- **Equations of motions:**

$$v(t) - R\,i(t) - \frac{L di(t)}{dt} - K_e\omega(t) = 0$$

$$\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = K_T i(t)$$

$$\omega(t) = \frac{d\theta(t)}{dt}$$

- Defining the state variable as:

$$x(t) = \begin{bmatrix}\theta(t) & \dot{\theta}(t) & i(t)\end{bmatrix}^T$$

- Input: Applied voltage

$$u(t) = v(t)$$

- Output: Motor positon

$$y(t) = \theta(t)$$

Example 3: DC Motor

- **Equations of motions:**

$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - K_e\frac{d\theta(t)}{dt} = 0$$

$$\frac{J_M d^2\theta(t)}{dt^2} + D_M\frac{d\theta(t)}{dt} = K_T i(t)$$

- Defining the state variable as:

$$x(t) = \begin{bmatrix}\theta(t) & \dot{\theta}(t) & i(t)\end{bmatrix}^T$$

- Input: Applied voltage

$$u(t) = v(t)$$

- Output: Motor positon

$$y(t) = \theta(t)$$

Write the system based on the state vector and output

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \frac{1}{J_M}(-D_M x_2 + K_T x_3)$$
$$\dot{x}_3 = \frac{1}{L}(-K_e x_2 - R x_3 + u)$$

$$x_1(t) = \theta(t)$$
$$x_2(t) = \dot{\theta}(t)$$
$$x_3(t) = i(t)$$

$$y(t) = x_1(t)$$

Example 3: State-Space of DC Motor

The system based on the state vector and output

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \frac{1}{J_M}(-D_M x_2 + K_T x_3)$$
$$\dot{x}_3 = \frac{1}{L}(-K_e x_2 - R x_3 + u)$$
$$y = x_1$$

- State: $x(t) = \begin{bmatrix}\theta(t) & \dot{\theta}(t) & i(t)\end{bmatrix}^T$
- Input: Applied voltage $u(t) = v(t)$
- Output: Motor positon $y(t) = \theta(t)$

- Expressing in terms of matric form:

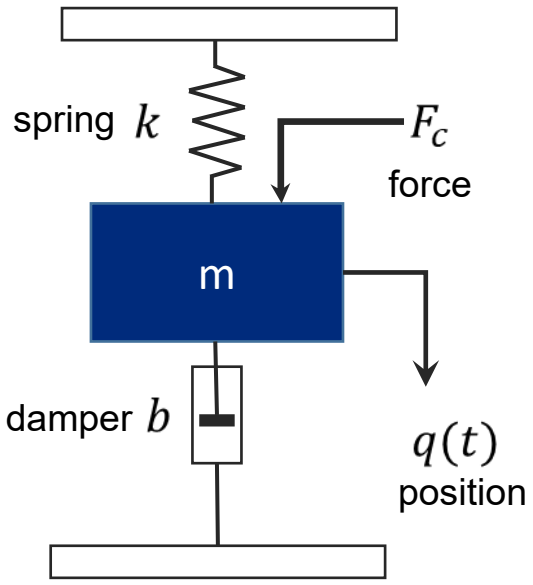
$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{D_M}{J_M} & \frac{K_T}{J_M} \\ 0 & -\frac{K_e}{L} & -\frac{R}{L} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}, \qquad C = [1 \quad 0 \quad 0], \qquad D = [0]$$

Exercise 1: Mass Spring Damper



Equation of motion for this system is:

$$m\ddot{q} + b\dot{q} + kq = F_c$$

Define the **states** as:

$$x_1 = q$$
$$x_2 = \dot{q}$$

Input is the force:

$$u = F_c$$

Output (i.e. measured variable) is position:

$$y = q$$

Rewrite as 1st order equations:

$$\dot{x}_1 = \dot{q}$$
$$\dot{x}_2 = \frac{1}{m}(-kq - b\dot{q}) + \frac{1}{m}F_c$$

Define w.r.t the system state

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \frac{1}{m}(-kx_1 - bx_2) + \frac{1}{m}u$$

The output equation is:

$$y = x_1$$

To get the matrix form:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

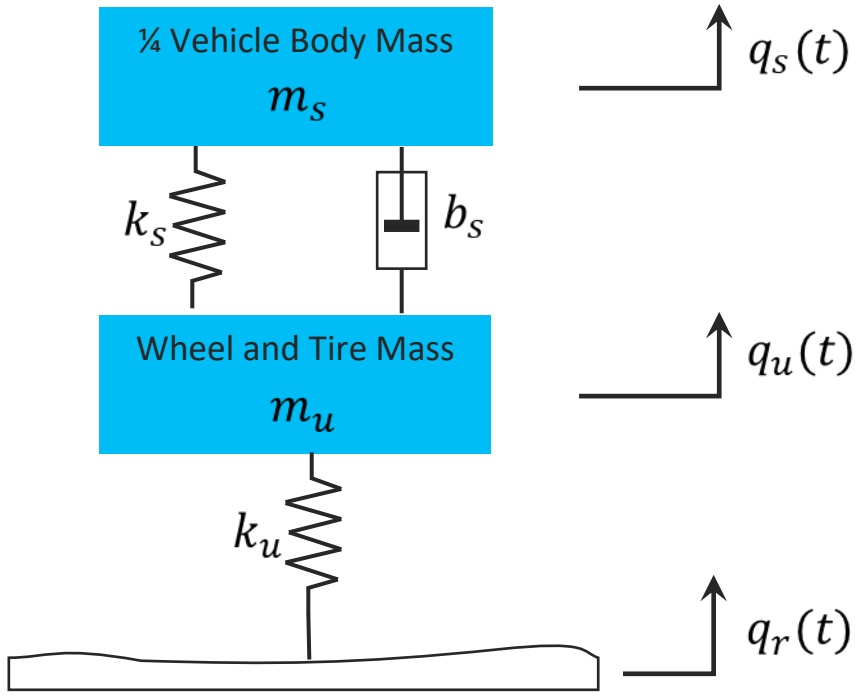
We have

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u$$

And

$$y = [1 \quad 0]x$$

Exercise 2: 2 DOF Suspension



Define the **states** as:

$$x_1 = q_u$$
$$x_2 = \dot{q}_u$$
$$x_3 = q_s$$
$$x_4 = \dot{q}_s$$

Input is the road:

$$u = \quad u_1 = q_r$$

Output (i.e. measured variable) is position:

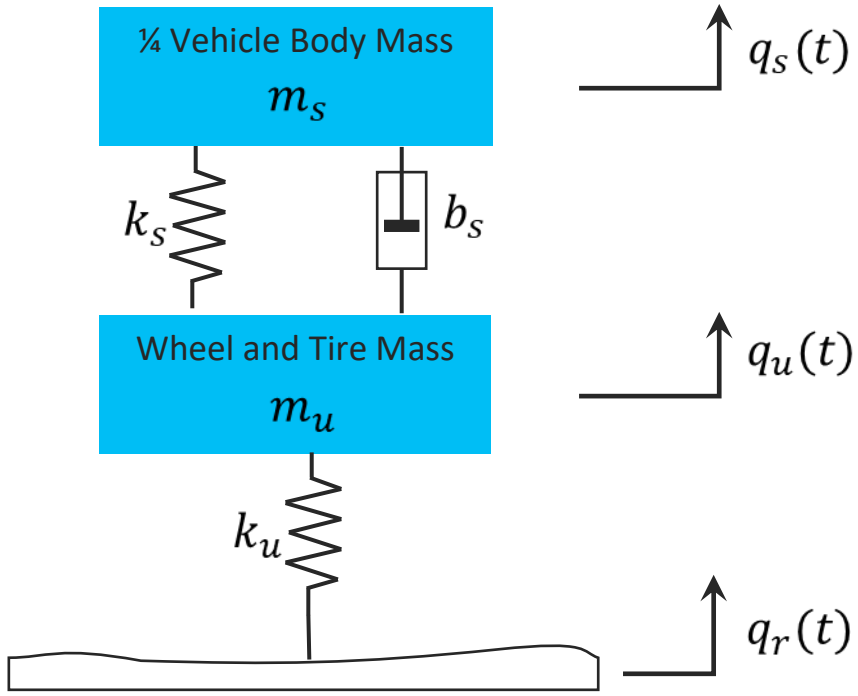
$$y_1 = q_s - q_u$$
$$y_2 = \ddot{q}_s$$

The equations of motion for this system are:

$$m_s\ddot{q}_s = -b_s(\dot{q}_s - \dot{q}_u) - k_s(q_s - q_u)$$
$$m_u\ddot{q}_u = b_s(\dot{q}_s - \dot{q}_u) + k_s(q_s - q_u) - k_u(q_u - q_r)$$

Note: The effect of gravity is eliminated from the EOMs by redefining the states w.r.t. to the eq. point.

Exercise 2: 2 DOF Suspension



The equations of motion for this system are:

$$m_s\ddot{q}_s = -b_s(\dot{q}_s - \dot{q}_u) - k_s(q_s - q_u)$$
$$m_u\ddot{q}_u = b_s(\dot{q}_s - \dot{q}_u) + k_s(q_s - q_u) - k_u(q_u - q_r)$$

Note: Effect of gravity is eliminated from the EOMs by redefining the states w.r.t. to the eq. point.

Substituting the state variables:

$$m_u\dot{x}_2 = b_s(x_4 - x_2) + k_s(x_3 - x_1) - k_u(x_1 - u_1)$$
$$m_s\dot{x}_4 = -b_s(x_4 - x_2) - k_s(x_3 - x_1)$$

Collecting the coefficients:

$$m_u\dot{x}_2 = -(k_u + k_s)x_1 - b_s x_2 + k_s x_3 + b_s x_4 + k_u u_1$$
$$m_s\dot{x}_4 = k_s x_1 + b_s x_2 - k_s x_3 - b_s x_4$$

Exercise 2: State-Space of 2 DOF Suspension

Given the EOMs w.r.t. to the state variables:

$$m_u\dot{x}_2 = -(k_u + k_s)x_1 - b_s x_2 + k_s x_3 + b_s x_4 + k_u u_1$$
$$m_s\dot{x}_4 = k_s x_1 + b_s x_2 - k_s x_3 - b_s x_4$$

$$y_1 = q_s - q_u$$
$$y_2 = \ddot{q}_s$$

Expressing this is terms of the matrix form:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

We have

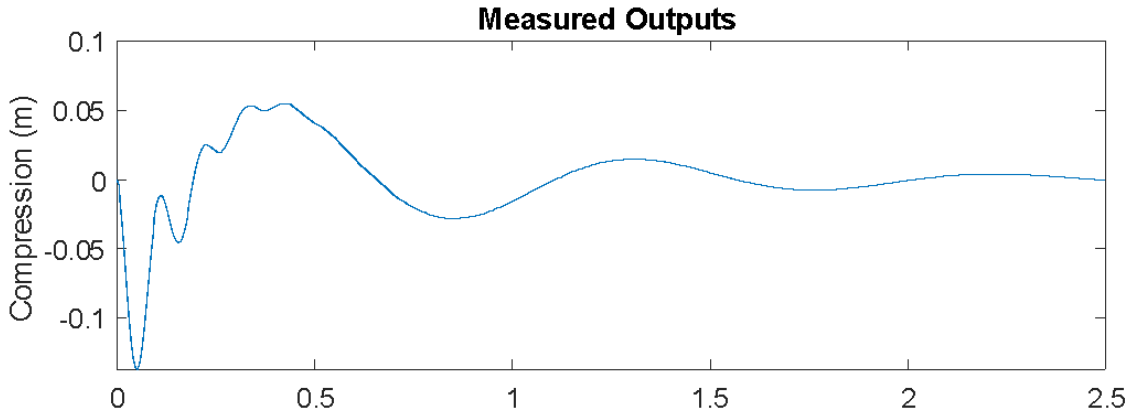
$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_s + k_u)/m_u & -b_s/m_u & k_s/m_u & b_s/m_u \\ 0 & 0 & 0 & 1 \\ k_s/m_s & b_s/m_s & -k_s/m_s & -b_s/m_s \end{bmatrix}$$

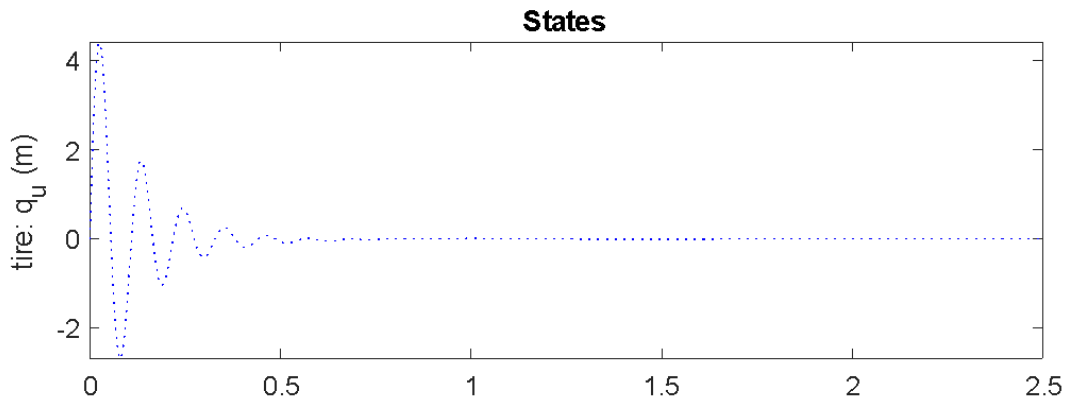
$$B = \begin{bmatrix} 0 \\ k_u/m_u \\ 0 \\ 0 \end{bmatrix}$$

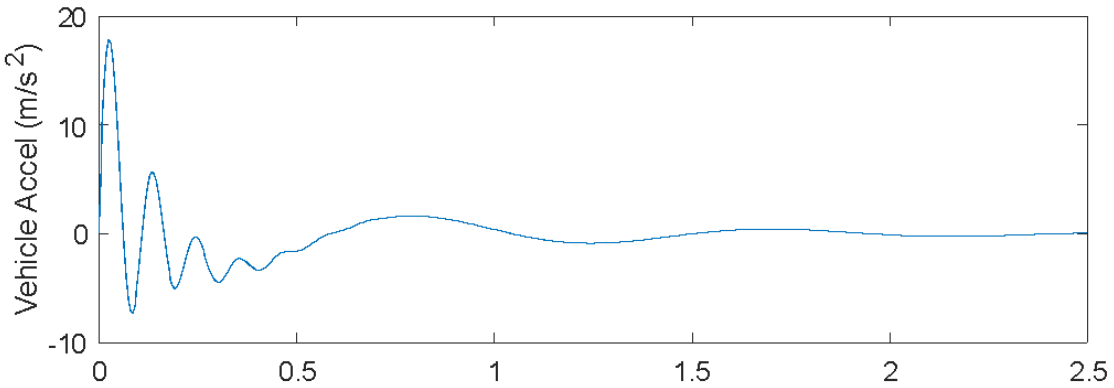
$$C = \begin{bmatrix} -1 & 0 & 1 & 0 \\ k_s/m_s & b_s/m_s & -k_s/m_s & -b_s/m_s \end{bmatrix}$$

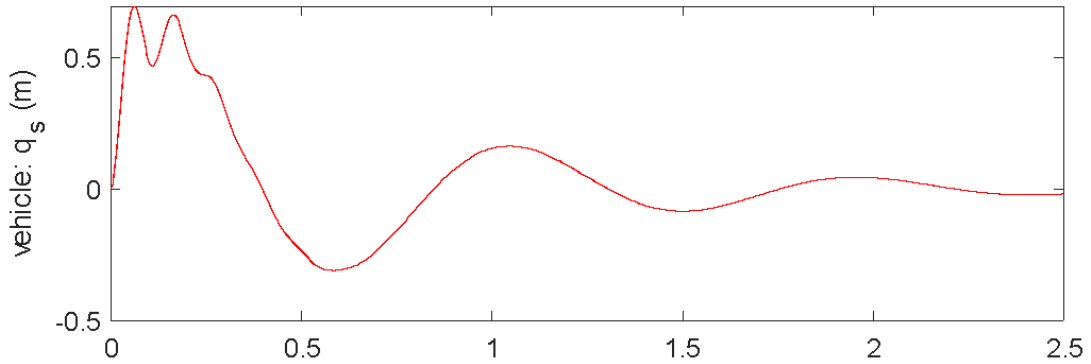
$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Response to a 1 N Step









State variables

- The internal state variables are the smallest possible subset of system variables that can represent the entire state of the system at any given time.
- The minimum number of state variables required to represent a given system, n, is usually equal to the order of the system's defining differential equation, but not necessarily.
- In electric circuits, the number of state variables is often, though not always, the same as the number of energy storage elements in the circuit such as capacitors and inductors.
- If the system is represented in transfer function form, the minimum number of state variables is equal to the order of the transfer function's denominator after it has been reduced to a proper fraction.

Convert Transfer Function to State-Space: Constant Numerator

Arrange the numerator and denominator coefficients of a transfer function into the state-space matrices.

$$y^{(n)} + a_{n-1}\,y^{(n-1)} + \cdots + a_1\dot{y} + a_0 y = b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} u$$

Define

$x_1 = y$

$x_2 = \dot{y}$

$x_3 = \ddot{y}$

$\vdots$

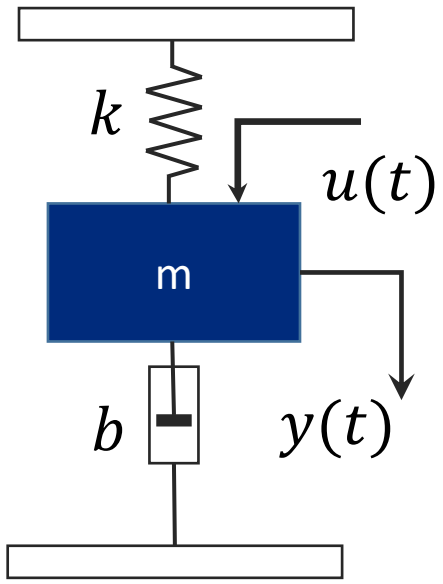
$x_{n-1} = y^{(n-2)}$

$x_n = y^{(n-1)}$

$$y = \begin{bmatrix} 1 & 0 & 0 \ldots 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \cdots 0 & 0 \end{bmatrix} x$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

Example: Converting TF to SS



EOM for this system is $m\ddot{y} + b\dot{y} + ky = u$

Transfer function

$$\frac{Y(s)}{U(s)} = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

The state-space form:

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u$$

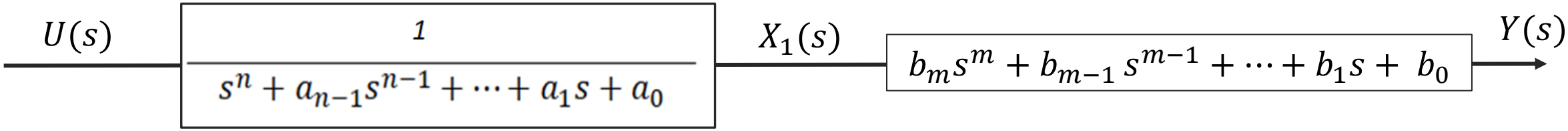
And

$$x_1 = y, \qquad x_2 = \dot{y}$$

$$y = [1 \quad 0]x$$

Convert TF to SS: general form

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1}\, u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u$$
$$(s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0) Y(s) = (b_m s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0) U(s)$$



$$\frac{Y(s)}{U(s)} = \frac{s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0}$$

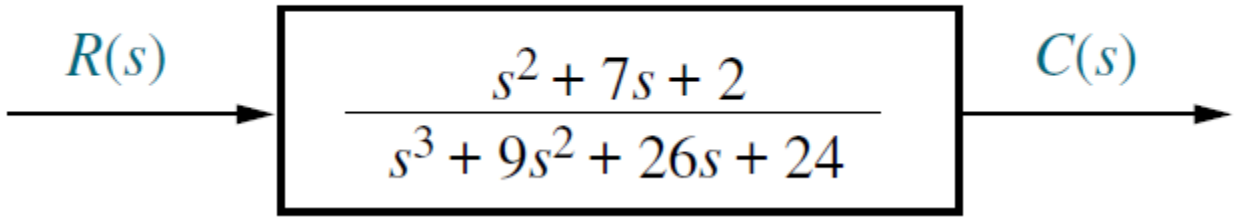
$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

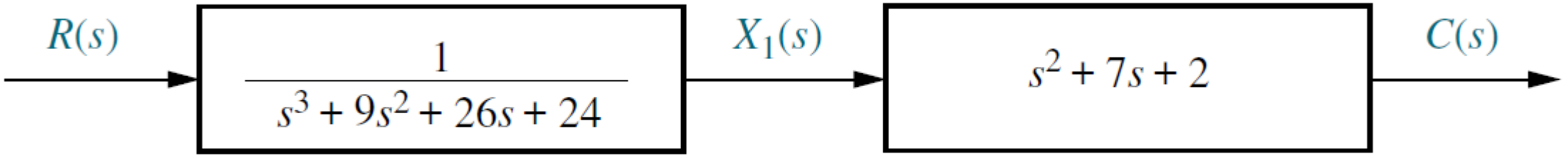
$$y = [b_0 \quad b_1 \quad \ldots \quad b_{m-1} \quad b_m]\, x$$

Internal state: $x(t) = \left[x_1 \quad \dot{x}_1 \quad \ldots \quad x^{(n-1)}\right]$

Example: From TF to SS

- Find the state-space representation of the transfer function

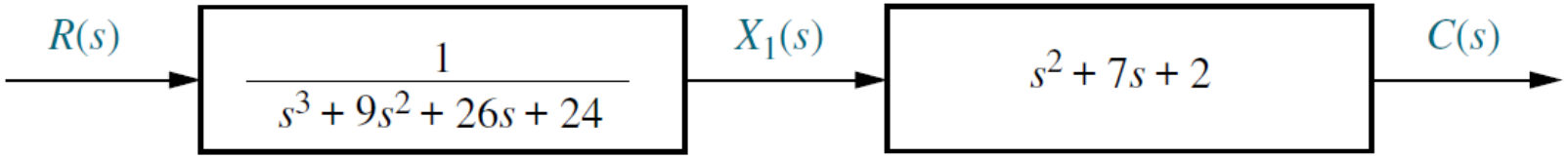




Internal variables:
$X_2(s),\ X_3(s)$

$$X_2(s) = sX_1(s)$$
$$X_3(s) = sX_2(s) = s^2X_1(s)$$

State vector: $X(s) = [X_1(s)\ \ X_2(s)\ \ X_3(s)]^T$
Output: $Y(s) = C(s)$



Internal variables:
$X_2(s),\ X_3(s)$

$$C(s) = (b_2 s^2 + b_1 s + b_0) X_1(s) = (s^2 + 7s + 2) X_1(s)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Convert State-Space to Transfer Function

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Take Laplace transform for the state space model:

$$sX(s) = AX(s) + BU(s)$$
$$Y(s) = CX(s) + DU(s)$$

Solving:

$$X(s) = (sI - A)^{-1}BU(s)$$

$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s) = [C(sI - A)^{-1}B + D]U(s)$$

If it is SISO system $\boxed{T(s) = C(sI - A)^{-1}B + D}$

Convert State-Space to Transfer Function

State-Space Model:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$x(n \times 1)$
$y\ (m \times 1)$
$u(r \times\ 1)$

$A_{n\times n}$
$B_{n\times r}$
$C_{m\times n}$
$D_{m\times r}$

Transfer functions:

$$T(s) = C(sI - A)^{-1}B + D = \begin{bmatrix} \frac{Y_1}{U_1} & \frac{Y_1}{U_2} & \cdots & \frac{Y_1}{U_r} \\ \frac{Y_2}{U_1} & \frac{Y_2}{U_2} & \cdots & \frac{Y_2}{U_r} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{Y_m}{U_1} & \frac{Y_m}{U_2} & \cdots & \frac{Y_m}{U_r} \end{bmatrix}_{m\times r}$$

Example: convert SS to TF

Problem: Converting the following state-space system to a transfer function

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t)$$

$$y = [1.5 \quad 0.625]x$$

Solution: $\boldsymbol{n = 2, m = 1, r = 1}$

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B = \frac{3s + 5}{s^2 + 4s + 6}$$

State-Space Modelling: Takeaways

Benefits

- Easier to model MIMO systems
- Once in state-space form, can use a variety of state-feedback control techniques

Drawbacks

- Not necessary for SISO and other simple systems, transfer functions modelling is as if not easier to use
- Transfer function based control design simpler and leads to desired results more quickly (less tuning).